

Max. Marks: 100

EC1002-1 – APPLIED DIGITAL LOGIC DESIGN

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Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which of the following logic gates provides output as 0 when both inputs are the same (either 0 or 1)?
A) XNOR
B) XOR
C) NOR
D) NAND
2. An XNOR gate produces an output only when the two inputs are:
A) High
B) Different
C) Low
D) Same
3. The octal equivalent of 1100101.001010 is
A) 624.12
B) 145.12
C) 154.12
D) 145.21
4. Convert the binary equivalent 10101 to its decimal equivalent.
A) 21
B) 12
C) 22
D) 31
5. Which of the following is not a binary number?
A) 111
B) 101
C) 11E
D) 000
6. What could be the maximum value of a single digit in an octal number system?
A) 8
B) 7
C) 6
D) 5
7. The maximum number of bits sufficient to represent an octal number in binary is
A) 4
B) 3
C) 7
D) 8
8. Convert $(22)_8$ into its corresponding decimal number
A) 28
B) 18
C) 81
D) 82
9. If A and B are the inputs of a half adder, the carry is given by
A) A AND B
B) A OR B
C) A XOR B
D) A EX-NOR B
10. Half-adders have a major limitation in that they cannot _____.
A) Accept a carry bit from a present stage
B) Accept a carry bit from a next stage
C) Accept a carry bit from a previous stage
D) Accept a carry bit from the following stages
11. If A, B and C are the inputs of a full adder then the carry is given by _____.
A) A AND B OR (A OR B) AND C
B) A OR B OR (A AND B) C
C) (A AND B) OR (A AND B) C
D) A XOR B XOR (A XOR B) AND C
12. How many AND, OR and EXOR gates are required for the configuration of full adder?
A) 2, 2, 2
B) 2, 1, 2
C) 3, 1, 2
D) 4, 0, 1
13. Let the input of a subtractor is A and B then what the output will be if $A = B$?
A) 0
B) 1
C) A
D) B
14. Let A and B is the input of a subtractor then the output will be
A) A XOR B
B) A AND B
C) A OR B
D) A EXNOR B
15. Full subtractor is used to perform subtraction of _____.
A) 2 bits
B) 3 bits
C) 4 bits
D) 8 bits

16. The full subtractor can be implemented using _____
 A) Two XOR and an OR gates B) Two half subtractors and an OR gate
 C) Two multiplexers and an AND gate D) Two comparators and an AND gate
17. The characteristic equation for J-K flip flop is _____
 A) $J\bar{Q} + \bar{K}Q$ B) $JQ + KQ$
 C) $\bar{J}Q + K\bar{Q}$ D) $J\bar{Q} + \bar{K}Q$
18. The minimum number of flip-flops that can be used to construct a modulus-5 counter is _____
 A) 3 B) 8
 C) 5 D) 10
19. A serial in parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains _____
 A) 0111 B) 0101
 C) 1110 D) 1100
20. What is meant by parallel-loading the register?
 A) Shifting the data in all flip-flops simultaneously B) Loading data in two of the flip-flops
 C) Loading data in all four flip-flops at the same time D) Momentarily disabling the synchronous SET and RESET inputs

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Design a combinational logic circuit which takes two, 2 bit numbers as its inputs and generates an output to indicate when sum of the two numbers is odd.	8	L3	1	1
b)	Implement the following function using NAND gates only. $P = x'y + xy' + z$	4	L3	2	1
c)	Find the minimal sum for the following expression. $F(w, x, y, z) = \sum m(1, 5, 7, 8, 9, 10, 11, 13, 15)$	4	L3	2	1
2. a)	With a neat logic diagram, truth table and expression sketch AND, OR, NAND and EXOR gates used in digital circuits.	8	L2	2	1
b)	Using K-Map, find Minimal Sum expression for the following switching function $f(w, x, y, z) = \sum m(1, 5, 6, 7, 9, 11, 12, 13) + dc(0, 3, 4)$	4	L3	2	1
c)	Perform number code conversion for the following and show each step clearly. i) $CF. 52_{(16)} = ?_{(10)}$ ii) $1376.185_{(10)} = ?_{(2)}$	4	L3	1	1
3. a)	Using QM method, simplify the following function $f(a, b, c, d) = \sum m(0, 1, 3, 6, 8, 9, 12, 13)$	8	L3	2	1
b)	Convert the following equation into Canonical SOP form $R(L, M, N) = L + \bar{M}(\bar{N}M + \bar{M}L)$	4	L3	2	1
c)	Define and give example to each. i) Minterm ii) Maxterm	4	L1	1	1
Unit – II					
4. a)	With a neat logic diagram, function table and timing diagram explain the operation of pulse triggered JK flip-flop.	8	L2	4	1
b)	Implement the following functions using appropriate decoder with minimum number of gate inputs. i) $F1(w, x, y) = \sum m(1, 3, 5, 6, 7)$ ii) $F2(w, x, y) = \prod M(1, 2, 3, 5, 6, 7)$	4	L3	3	1
c)	Differentiate between i) Combinational Logic circuits and sequential logic circuits ii) Latch and Flip flop	4	L2	4	1

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|-------|---|---|----|---|---|
| 5. a) | Define Half and Full Adder. Design the circuit of Half and Full Adder with truth table and write the logic diagram. | 8 | L3 | 3 | 1 |
| b) | Realize the following Boolean function using an appropriate multiplexer and external gates with a, b as select lines. | | | | |
| | $f(a,b,c) = \sum m(0,1,5,6,7)$ | 4 | L3 | 3 | 1 |
| c) | What are the characteristic equations of Flip flop? Obtain the characteristic equation of JK Flip Flop | 4 | L2 | 4 | 1 |
| 6. a) | With neat logic diagram and function table explain the positive edge triggered D Flip flop and also write its timing diagram. | 8 | L2 | 4 | 1 |
| b) | Design a combinational logic circuit to compare the two, 1 bit numbers. | 4 | L3 | 3 | 1 |
| c) | Explain the operation of 4-bit parallel adder using full adders with an example. | 4 | L2 | 3 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Design a synchronous counter to count $\rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 6 \rightarrow 0$ using positive edge triggered D flip flops with minimal combinational gating. | 8 | L3 | 5 | 1 |
| b) | What is a shift register? With the help of a suitable example explain the following operations: | | | | |
| | i) SIPO | | | | |
| | ii) SISO | 8 | L2 | 5 | 1 |
| 8. a) | What is a Ripple counter? Explain the working of 3-bit ripple down counter with the help of a logic diagram, timing diagram and counting sequence using negative edge triggered T Flip Flops. | 8 | L2 | 5 | 1 |
| b) | Convert JK Flip flop to | | | | |
| | i) D Flip flop | | | | |
| | ii) T Flip flop | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
